TRIPATHI STUDY ZONE

JEE MAIN | NEET | AIIMS | KVPY | BITSAT

Mock Test 2

Time : 3 hrs. **Max. Marks : 300**

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three parts in the question paper consisting of Physics (Q.no.1 to 30), Chemistry (Q.no.31 to 60) and Mathematics (Q. no.61 to 90). Each part is divided into two sections, Section A consists of 20 multiple choice questions & Section B consists of 10 Numerical value answer Questions.
   In Section B, candidates have to attempt **only 5 questions out of 10**.
5. There will be only one correct choice in the given four choices in Section A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for unattempted question. For Section B 4 marks will be awarded for correct answer and zero for marked for each review / unattempted/incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

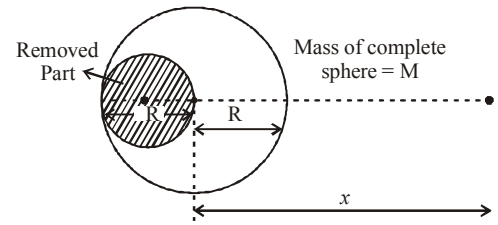
Section - A

1. N divisions on the main scale of a vernier callipers coincide with N + 1 divisions on the vernier scale. If each division on the main scale is of 'a' units, the least count of the instrument is

   (1) $\frac{a}{N+1}$ (2) $\frac{a}{N-1}$

   (3) $\frac{2a}{N-1}$ (4) $\frac{2a}{N+1}$

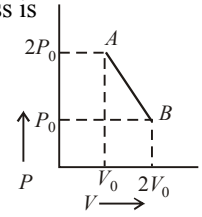
2. The gravitational field, due to the 'left over part' of a uniform sphere (from which a part as shown, has been 'removed out'), at a very far off point, P, located as shown, would be (nearly) :



   (1) $\frac{5}{6}\frac{GM}{x^2}$ (2) $\frac{8}{9}\frac{GM}{x^2}$

   (3) $\frac{7}{8}\frac{GM}{x^2}$ (4) $\frac{6}{7}\frac{GM}{x^2}$

3. n moles of an ideal gas undergo a process $A \rightarrow B$ as shown in the figure. Maximum temperature of the gas during the process is



(1) $\frac{9P_0V_0}{nR}$ (2) $\frac{3P_0V_0}{2nR}$

(3) $\frac{9P_0V_0}{2nR}$ (4) $\frac{9P_0V_0}{4nR}$

4. Which of these nuclear reactions is possible –

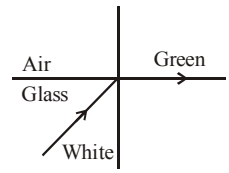
(1) ${}^{23}_{11}\text{Na} + {}^{1}_{1}\text{H} \longrightarrow {}^{20}_{10}\text{Ne} + {}^{4}_{2}\text{He}$

(2) ${}^{10}_{5}\text{B} + {}^{4}_{2}\text{He} \longrightarrow {}^{13}_{7}\text{N} + {}^{1}_{1}\text{H}$

(3) ${}^{10}_{5}\text{B} + {}^{1}_{0}\text{n} \longrightarrow {}^{11}_{5}\text{B} + \beta^- + \bar{\nu}$

(4) ${}^{14}_{7}\text{N} + {}^{1}_{1}\text{H} \longrightarrow {}^{12}_{6}\text{C} + \beta^- + \bar{\nu}$

5. White light is incident on the interface of glass and air as shown in the figure. If green light is just totally internally reflected then the emerging ray in air contains



(1) yellow, orange, red

(2) violet, indigo, blue

(3) All colours

(4) All colours except green

6. What is the conductivity of a semiconductor sample having electron concentration of 5×10^{18} m^{-3}, hole concentration of 5×10^{19} m^{-3}, electron mobility of 2.0 $m^2 V^{-1} s^{-1}$ and hole mobility of 0.01 $m^2 V^{-1} s^{-1}$?

(Take charge of electron as 1.6×10^{-19} C)

(1) $1.68\ (\Omega - m)^{-1}$ (2) $1.83\ (\Omega - m)^{-1}$

(3) $0.59\ (\Omega - m)^{-1}$ (4) $1.20\ (\Omega - m)^{-1}$

7. A body is executing simple harmonic motion. At a displacement x from mean position, its potential energy is $E_1 = 2J$ and at a displacement y from mean position, its potential energy is $E_2 = 8J$. The potential energy E at a displacement (x + y) from mean position is

(1) 10J (2) 14J

(3) 18J (4) 4J

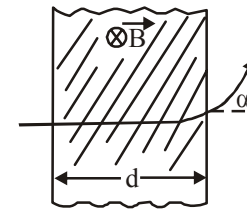
8. A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field B. The field occupies a region of space by width 'd'. If α be the angle of deviation of proton from initial direction of motion (see figure), the value of $sin\ \alpha$ will be :

(1) $qV\sqrt{\frac{Bd}{2m}}$

(2) $\frac{B}{2}\sqrt{\frac{qd}{mV}}$

(3) $\frac{B}{d}\sqrt{\frac{q}{2mV}}$

(4) $Bd\sqrt{\frac{q}{2mV}}$



9. Two wires are made of the same material and have the same volume. However wire 1 has cross-sectional area A and wire 2 has cross-sectional area $3A$. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount?

(1) $4F$ (2) $6F$ (3) $9F$ (4) F

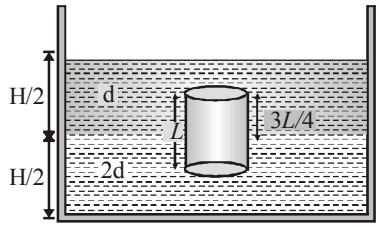
10. A homogeneous solid cylinder of length L (L < H/2). Cross-sectional area A/5 is immersed such that it floats with its axis vertical at the liquid-liquid interface with length L/4 in the denser liquid as shown in the fig. The lower density liquid is open to atmosphere having pressure P_0. Then density of solid is given by

(1) $\frac{5}{4}d$

(2) $\frac{4}{5}d$

(3) d

(4) $\frac{d}{5}$



11. In a Young's double slit experiment with light of wavelength λ, fringe pattern on the screen has fringe width β. When two thin transparent glass (refractive index μ) plates of thickness t_1 and t_2 ($t_1 > t_2$) are placed in the path of the two beams respectively, the fringe pattern will shift by a distance

(1) $\frac{\beta(\mu-1)}{\lambda}\left(\frac{t_1}{t_2}\right)$ (2) $\frac{\mu\beta}{\lambda}\frac{t_1}{t_2}$

(3) $\frac{\beta(\mu-1)}{\lambda}(t_1-t_2)$ (4) $(\mu-1)\frac{\lambda}{\beta}(t_1+t_2)$

12. A dipole consisting of two charges +q and –q joined by a massless rod of length ℓ, is seen oscillating with a small amplitude in a uniform electric field of magnitude E. The period of oscillation

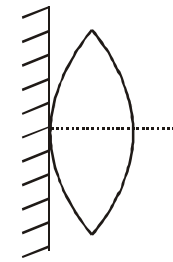
(1) is proportional to E

(2) is proportional to 1/E

(3) is $\pi\sqrt{\frac{m\ell}{3qE}}$

(4) is proportional to $\frac{1}{\sqrt{E}}$ but $\neq \pi\sqrt{\frac{m\ell}{3qE}}$

13. A thin convex lens of focal length 'f' is put on a plane mirror as shown in the figure. When an object is kept at a distance 'a' from the lens - mirror combination, its image is formed at a distance $\frac{a}{3}$ in front of the combination. The value of 'a' is :



(1) 3f

(2) $\frac{3}{2}$f

(3) f

(4) 2f

14. A sinusoidal wave is propagating in negative x-direction in a string stretched along x-axis. A particle of string at x = 2m is found at its mean position and it is moving in positive y-direction at t = 1 sec. The amplitude of the wave, the wavelength and the angular frequency of the wave are 0.1 meter, π/2 meter and 2π rad/sec respectively.

The equation of the wave is

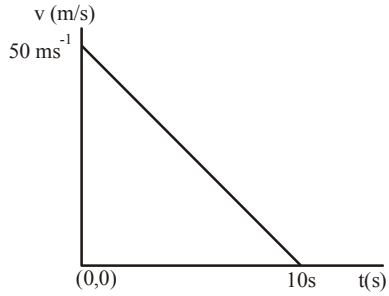
(1) $y = 0.1 \sin(4\pi(t-1)+8(x-2))$

(2) $y = 0.1 \sin((t-1)-(x-2))$

(3) $y = 0.1 \sin(2\pi(t-1)+4(x-2))$

(4) None of these

15. Velocity–time graph for a body of mass 10 kg is shown in figure. Work–done on the body in first two seconds of the motion is :



(1) – 9300 J (2) 12000 J
(3) –4500 J (4) –12000 J

16. A charged particle of mass m and having a charge Q is placed in an electric field E which varies with time as $E = E_0 \sin \omega t$. What is the amplitude of the S.H.M. executed by the particle?

(1) $\frac{QE_0}{m\omega^2}$ (2) $\frac{1}{2}\frac{QE_0}{m\omega^2}$

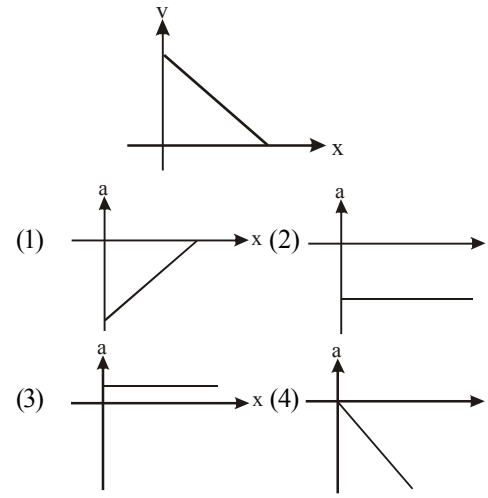
(3) $\frac{2QE_0}{m\omega^2}$ (4) None of these

17. In a photoelectric experiment, with light of wavelength λ, the fastest electron has speed v. If the exciting wavelength is changed to $3\lambda/4$, the speed of the fastest emitted electron will become –

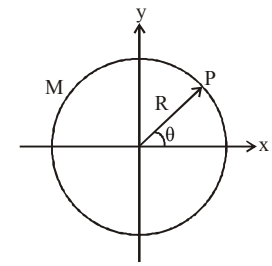
(1) $v\sqrt{\frac{3}{4}}$ (2) $v\sqrt{\frac{4}{3}}$

(3) less than $v\sqrt{\frac{4}{3}}$ (4) greater than $v\sqrt{\frac{4}{3}}$

18. A particle moves along x-axis with initial position x = 0. Its velocity varies with x-coordinate as shown in graph. The acceleration 'a' of this particle varies with x as –



19. A ring of mass M and radius R lies in x-y plane with its centre at origin as shown. The mass distribution of ring is non-uniform such that at any point P on the ring, the mass per unit length is given by $\lambda = \lambda_0 \cos^2\theta$ (where λ_0 is a positive constant). Then the moment of inertia of the ring about z-axis is –



(1) MR^2

(2) $\frac{1}{2}MR^2$

(3) $\frac{1}{2}\frac{M}{\lambda_0}R$

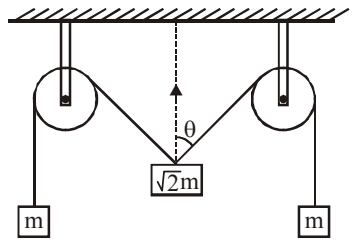
(4) $\frac{1}{\pi}\frac{M}{\lambda_0}R$

20. The electric potential V is given as a function of distance x by $V = (5x^2 + 10x - 9)$ volt. Value of electric field at x = 1m is

(1) –20 V/m (2) 6 V/m
(3) 11 V/m (4) 20 V/m

Section - B

21. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be _____ degree.



22. A current of 4A produces a deflection of 30° in the galvanometer. The figure of merit is _____ A/rad.

23. A uniform thin rod AB of length L has linear mass density $\mu(x) = a + \frac{bx}{L}$, where x is measured from A. If the CM of the rod lies at a distance of $\left(\frac{7}{12}\right)$L from A, then a and b are related as b = xa. Find the value of x.

24. A transmitting antenna at the top of a tower has height 32 m and height of the receiving antenna is 50 m. What is the maximum distance (in km) between them for satisfactory communication in line of sight (LOS) mode?

25. A ball of mass 160 g is thrown up at an angle of 60° to the horizontal at a speed of 10 ms^{-1}. The angular momentum of the ball at the highest point of the trajectory with respect to the point from which the ball is thrown is _____ kg m^2/s. ($g = 10\ ms^{-2}$)

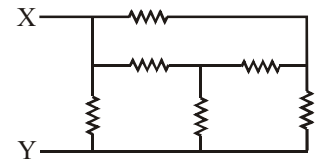
26. A series LR circuit is connected to an ac source of frequency ω and the inductive reactance is equal to 2R. A capacitance of capacitive reactance equal to R is added in series with L and R. The ratio of the new power factor to the old one is $\sqrt{\frac{x}{2}}$. Find the value of x.

27. A mass of 50g of water in a closed vessel, with surroundings at a constant temperature takes 2 minutes to cool from 30°C to 25°C. A mass of 100g of another liquid in an identical vessel with identical surroundings takes the same time to cool from 30° C to 25° C. The specific heat of the liquid is _____ kcal/kg.
(The water equivalent of the vessel is 30g.)

28. The magnetic field in a travelling electromagnetic wave has a peak value of 20 nT. The peak value of electric field strength is _____ V/m.

29. An ideal monatomic gas with pressure P, volume V and temperature T is expanded isothermally to a volume $2V$ and a final pressure P_i. If the same gas is expanded adiabatically to a volume $2V$, the final pressure is P_a. The ratio $\frac{P_a}{P_i}$ is _____.

30. Six resistors of 10Ω each are connected as shown. The equivalent resistance between points X and Y is _____.



CHEMISTRY

Section - A

31. Consider the following reactions

(i) $Cd^{2+}(aq) + 2e^- \longrightarrow Cd(s)$, $E^\circ = -0.40\,V$

(ii) $Ag^+(aq) + e^- \longrightarrow Ag(s)$, $E^\circ = 0.80\,V$

For the galvanic cell involving the above reactions. Which of the following is not correct?

(1) Molar concentration of the cation in the cathodic compartment changes faster than that of the cation in the anodic compartment.

(2) E_{cell} increase when Cd^{2+} solution is diluted.

(3) Twice as many electrons pass through the cadmium electrode as through silver electrode.

(4) E_{cell} decreases when Ag^+ solution is diluted.

32. A container contains certain gas of mass 'm' of high pressure. Some of the gas has been allowed to escape from the container and after some time the pressure of the gas becomes half and its absolute temperature 2/3 rd. The amount of the gas escaped is

(1) 2/3 m (2) 1/2 m

(3) 1/4 m (4) 1/6 m

33. The hybrid state of S in SO_3 is similar to that of

(1) C in C_2H_2 (2) C in C_2H_4

(3) C in CH_4 (4) C in CO_2

34. Choose the correct options –

(1) Tranquillizers are called psychotherapeutic drugs.

(2) Vitamin C, Vitamin E and β-carotene are antioxidants.

(3) Sodium or potassium salt of a long chain fatty acid is called soap.

(4) All of these.

35. Electrode potential of the half cell Pt (s) | Hg (l) | Hg_2Cl_2(s) | Cl^- (aq) can be increased by :

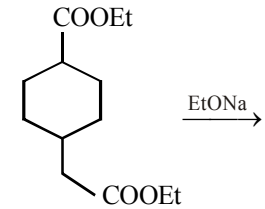
(1) Increasing $[Cl^-]$

(2) Decreasing $[Cl^-]$

(3) Increasing Hg_2Cl_2 (s)

(4) Decreasing Hg (l)

36. What is the name of the following reaction :



(1) Knoevenagel reaction

(2) Perkin reaction

(3) Reformatsky reaction

(4) Dieckmann's condensation.

37. Choose the correct options –

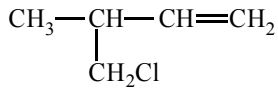
(1) The change of atmospheric temperature with altitude is called the lapse rate.

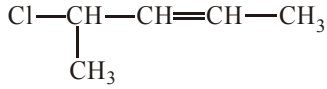
(2) The gases responsible for greenhouse effects are CO_2, water vapour, CH_4 and ozone.

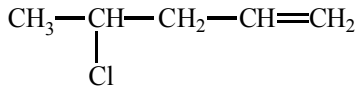
(3) The order of contribution to the acid rain is $H_2SO_4 < HNO_3 < HCl$

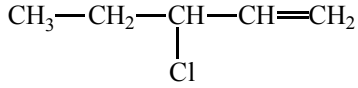
(4) Both (1) and (2)

38. Optically active isomer (A) of (C_5H_9Cl) on treatment with one mole of H_2 gives an optically inactive compound (B) compound (A) will be:

(1) 

(2) 

(3) 

(4) 

39. Which statement is/are correct about ICl_3 molecule ?
(1) All I – Cl bonds are equivalent.
(2) Molecule is polar and non-planar.
(3) All adjacent bond angles are equal.
(4) All hybrid orbitals of central atom having equal *s*-character.

40. Isopropyl alcohol is obtained by reacting which of the following alkenes with concentrated H_2SO_4 followed by boiling with H_2O?
(1) Ethylene (2) Propylene
(3) 2-Methylpropene (4) Isoprene

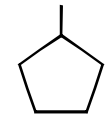
41. In electro-refining of metal the impure metal is used to make the anode and a strip of pure metal as the cathode, during the electrolysis of an aqueous solution of a complex metal salt. This method cannot be used for refining of
(1) silver (2) copper
(3) aluminium (4) sodium

42. Which of the following compounds does not give nitrogen gas on heating?
(1) NH_4NO_2 (2) $(NH_4)_2SO_4$
(3) NH_4ClO_4 (4) $(NH_4)_2Cr_2O_7$

43. Flocculation value of $BaCl_2$ is much less than that of KCl for sol A and flocculation value of Na_2SO_4 is much less than that of NaBr for sol B. The correct statement among the following is :
(1) Both the sols A and B are negatively charged.
(2) Sol A is positively charged and Sol B is negatively charged.
(3) Both the sols A and B are positively charged.
(4) Sol A is negatively charged and sol B is positively charged.

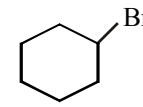
44. $HC \equiv CH \xrightarrow[H_2SO_4]{HgSO_4} (A) \xrightarrow[5\,^oC]{dil.\,OH^-} (B)$

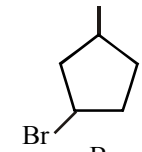
Give the IUPAC name of "B" is –
(1) 2-butanal
(2) 3-hydroxy butanal
(3) 3-formyl-2-propanol
(4) 4-oxo-2-propanol

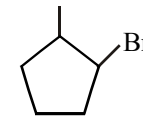
45.  $\xrightarrow{Br_{2/h\nu}}$ Major (X) $\xrightarrow[KOH/\Delta]{Alcoholic}$

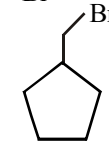
Major (Y) $\xrightarrow[Peroxide]{H-Br}$ Major (Z)

Major final product (Z) is –

(1) 

(2) 

(3) 

(4) 

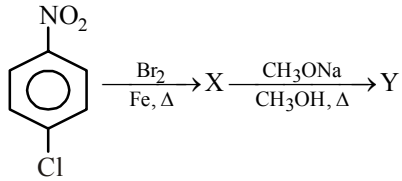
46. Choose the incorrect option –

(1) When a dilute solution of an acid is added to a dilute solution of a base, neutralization reaction takes place.

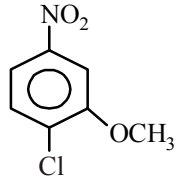
(2) In acid-base titrations, at the end point, the amount of acid becomes chemically equivalent to the amount of base present.

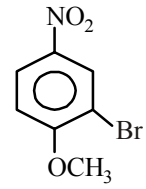
(3) In the case of a strong acid and a strong base titration, at the end point the solution becomes acidic.

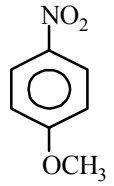
(4) In acid-base titrations the end point is determined by the hydrogen ion concentration of the solution.

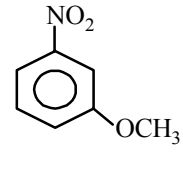
47. 

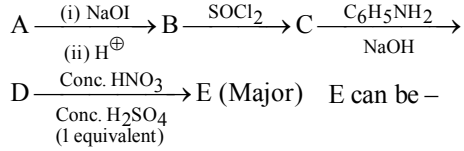
Product (Y) of this reaction is –

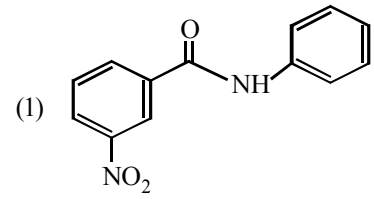
(1) 

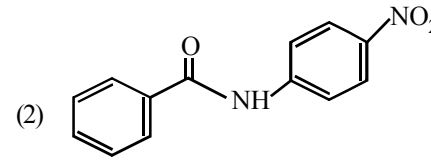
(2) 

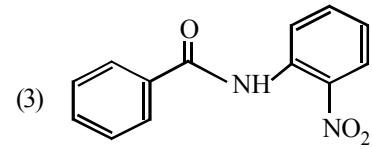
(3) 

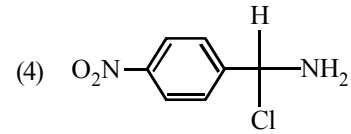
(4) 

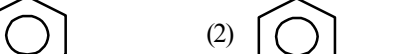
48. 

(1) 

(2) 

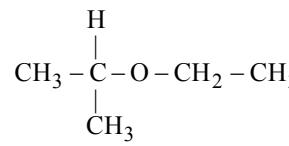
(3) 

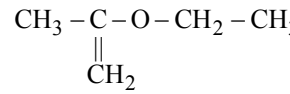
(4) 

49. 

the Iodoform test). The compound X is –

(1) $CH_3-CH=CH-O-CH_2-CH_3$

(2) 

(3) 

(4) Both (1) and (3)

50. If the elevation in boiling point of a solution of non-volatile, non-electrolytic and non-associating solute in a solvent ($K_b = x\ K\ kg\ mol^{-1}$) is y K, then the depression in freezing point of solution of same concentration would be (K_f of the solvent = z K kg mol^{-1})

(1) $\frac{2xz}{y}K$ (2) $\frac{yz}{x}K$

(3) $\frac{xz}{y}K$ (4) $\frac{yz}{2x}K$

Section - B

51. The density of CaF_2 (fluorite structure) is 3.18 g/cm^3. Find the length of the side of the unit cell is _____.

52. When the following aldohexose exists in its D-configuration, the total number of stereoisomers in its pyranose form is _____.

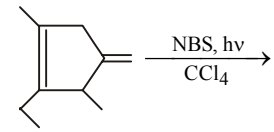
$CHO — CH_2 — CHOH — CHOH — CHOH — CH_2OH$

53. A 25.0 mm × 40.0 mm piece of gold foil is 0.25 mm thick. The density of gold is 19.32 g/cm^3. How many gold atoms are in the sheet in terms of $x \times 10^{22}$? (Atomic weight : Au = 197.0)

54. In a chemical reaction *A* is converted into *B*. The rates of reaction, starting with initial concentrations of *A* as 2×10^{-3} M and 1×10^{-3} M, are equal to 2.40×10^{-4} Ms^{-1} and 0.60×10^{-4} Ms^{-1} respectively. Find the order of reaction with respect to reactant *A*.

55. Rate constant k = 2.303 min^{-1} for a particular reaction. The initial concentration of the reactant is 1 mol/litre then calculatle the rate of reaction after 1 minute.

56. How many free radicals can be produced during following reaction (ignoring resonating structure) ?



57. An electron has a speed of 30,000 cm sec^{-1} accurate upto 0.001%. What is the uncertainty (in cm) in locating it's position?

58. The standard enthalpy of formation of NH_3 is – 46.0 kJ/mol. If the enthalpy of formation of H_2 from its atoms is – 436 kJ/mol and that of N_2 is – 712 kJ/mol, calculate the average bond enthalpy of N - H bond in NH_3.

59. The rate coefficient (k) for a particular reaction is 1.3×10^{-4} M^{-1} s^{-1} at 100 °C and 1.3×10^{-3} M^{-1} s^{-1} at 150 °C. What is the energy of activation (E_A) (in kJ/mol) for this reaction? (R = molar gas constant = 8.314 JK^{-1} mol^{-1})

60. The e.m.f. of the cell $Zn \mid Zn^{2+}$ (0.01M) $\mid\mid Fe^{2+}$ (0.001M) $\mid$ Fe at 298 K is 0.2905 then the value of equilibrium constant for the cell reaction 10^x. Find *x*.

MATHEMATICS

Section - A

61. Given $f(x)=\begin{cases}\sqrt{10-x^2} & \text{if } -3<x<3\\ 2-e^{x-3} & \text{if } x\geq 3\end{cases}$

The graph of f(x) is –

(1) continuous and differentiable at x = 3

(2) continuous but not differentiable at x = 3

(3) differentiable but not continuous at x = 3

(4) neither differentiable nor continuous at x = 3

62. The greatest and the least absolute value of $z+1$, where $|z+4|\leq 3$ are respectively

(1) 6 and 0 (2) 10 and 6

(3) 4 and 3 (4) None of these

63. The equation $(5x-1)^2+(5y-2)^2=(\lambda^2-4\lambda+4)(3x+4y-1)^2$ represents an ellipse if $\lambda \in$

(1) (0, 1] (2) (–1, 2)

(3) (2, 3) (4) (–1, 0)

64. If (– 4, 5) is one vertex and $7x-y+8=0$ is one diagonal of a square, then the equation of second diagonal is

(1) $x+3y=21$ (2) $2x-3y=7$

(3) $x+7y=31$ (4) $2x+3y=21$

65. Which of the following is always true?

(1) $(\sim p\Rightarrow q)=\sim q\Rightarrow\sim p$

(2) $(\sim p\vee q)\equiv\vee p\vee\sim q$

(3) $\sim(p\Rightarrow q)\equiv p\wedge\sim q$

(4) $\sim(p\vee q)\equiv\sim p\wedge\sim q$

66. Find $\int e^{\sin x}\left(\frac{x\cos^3 x-\sin x}{\cos^2 x}\right)dx$

(1) $x\,e^{\sin x}-e^{\sin x}\sec x+C$

(2) $x\,e^{\cos x}-e^{\sin x}\sec x+C$

(3) $x^2\,e^{\sin x}+e^{\sin x}\sec x+C$

(4) $2x\,e^{\sin x}-e^{\sin x}\tan x+C$

67. The function $f:[2,\infty)\to(0,\infty)$ defined by $f(x)=x^2-4x+a$, then the set of values of 'a' for which f(x) becomes onto is

(1) $(4,\infty)$ (2) $[4,\infty)$

(3) $\{4\}$ (4) ϕ

68. If α and β are the real roots of the equation $x^2-(k-2)x+(k^2+3k+5)=0$ $(k\in R)$.

Find the maximum and minimum values of $(\alpha^2+\beta^2)$.

(1) 18, 50/9 (2) 18, 25/9

(3) 27, 50/9 (4) None of these

69. The sum of the coefficient of all the terms in the expansion of $(2x-y+z)^{20}$ in which y do not appear at all while x appears in even powers and z appears in odd powers is –

(1) 0 (2) $\frac{2^{20}-1}{2}$

(3) 2^{19} (4) $\frac{3^{20}-1}{2}$

70. All the five digit numbers in which each successive digit exceeds is predecessor are arranged in the increasing order. The $(105)^{th}$ number does not contain the digit

(1) 1 (2) 2

(3) 6 (4) All of these

71. If $\hat{a}, \hat{b}, \hat{c}$ are non-coplanar unit vectors such that $\hat{a} \times (\hat{b} \times \hat{c}) = \frac{1}{\sqrt{2}}(\hat{b} + \hat{c})$ then the angle between the vectors $\hat{a}, \hat{b}$ is

(1) $3\pi/4$ (2) $\pi/4$

(3) $\pi/8$ (4) $\pi/2$

72. The greatest and the least values of $|z_1 + z_2|$ if $z_1 = 24 + 7i$ and $|z_2| = 6$ respectively are

(1) 31, 19 (2) 19, 31

(3) –19, –31 (4) –31, –19

73. Let P = (–1, 0), Q = (0, 0) and R = $(3, 3\sqrt{3})$ be three points. The equation of the bisector of the angle PQR is

(1) $\frac{\sqrt{3}}{2}x + y = 0$ (2) $x + \sqrt{3y} = 0$

(3) $\sqrt{3}x + y = 0$ (4) $x + \frac{\sqrt{3}}{2}y = 0$.

74. If $\int \sqrt{2}\sqrt{1+\sin x}dx = -4\cos(ax+b) + C$ then the value of (a, b) is :

(1) $\frac{1}{2}, \frac{\pi}{4}$ (2) $1, \frac{\pi}{2}$

(3) 1, 1 (4) None of these

75. If the tangent at any point on the curve $x^4 + y^4 = a^4$ cuts off intercepts p and q on the co-ordinate axes then the value of $p^{-4/3} + q^{-4/3}$ is

(1) $a^{-4/3}$ (2) $a^{-1/2}$

(3) $a^{1/2}$ (4) None

76. The area bounded by the curve $y^2(2a - x) = x^3$ and the line $x = 2a$ is

(1) $3\pi a^2$ sq. unit (2) $\frac{3\pi a^2}{2}$ sq. unit

(3) $\frac{3\pi a^2}{4}$ sq. unit (4) $\frac{6\pi a^2}{5}$ sq. unit

77. The expression satisfying the differential equation $(x^2 - 1)\frac{dy}{dx} + 2xy = 1$ is

(1) $x^2y - xy^2 = c$

(2) $(y^2 - 1)x = y + c$

(3) $(x^2 - 1)y = x + c$

(4) None of these

78. Let $f: R \to R$ and $f_n(x) = f(f_{n-1}(x)) \; \forall \; n \geq 2, n \in N$, the roots of equation $f_3(x)\, f_2(x)\, f(x) - 25 f_2(x)\, f(x) + 175\, f(x) = 375$, which also satisfy equation $f(x) = x$ will be

(1) 5 (2) 15

(3) 10 (4) Both (1) and (2)

79. The value of

$$\int_0^{\sin^2 x} \sin^{-1}\sqrt{t}\ dt + \int_0^{\cos^2 x} \cos^{-1}\sqrt{t}\ dt \text{ is}$$

(1) π (2) $\frac{\pi}{2}$

(3) $\frac{\pi}{4}$ (4) 1

80. If a is real and $\sqrt{2}ax + \sin By + \cos Bz = 0$,

$x + \cos By + \sin Bz = 0$, $-x + \sin By - \cos Bz = 0$, then the set of all values of a for which the system of linear equations has a non-trivial solution, is–

(1) $[1, 2]$ (2) $[-1, 1]$

(3) $[1, \infty]$ (4) $[2^{-1/2}, 2^{1/2}]$

Section - B

81. If two vertical poles 20 m and 80 m high stand apart on a horizontal plane, then the height (in m) of the point of intersection of the lines joining the top of each pole to the foot of other is _____.

82. If $x + iy = \frac{3}{\cos\theta + i\sin\theta + 2}$ then the value of $4x - x^2 - y^2$ is _____.

83. Box contains 2 one rupee, 2 five rupee, 2 ten rupee and 2 twenty rupee coin. Two coins are drawn at random simultaneously. The probability that their sum is Rs. 20 or more, is _____.

84. The value of the definite integral,

$\int_{\theta_1}^{\theta_2} \frac{d\theta}{1+\tan\theta} = \frac{501\pi}{K}$ where

$\theta_2 = \frac{1003\pi}{2008}$ and $\theta_1 = \frac{\pi}{2008}$. The value of K is _____.

85. The expansion of $(1 + x)^n$ has 3 consecutive terms with coefficients in the ratio 1 : 2 : 3 and can be written in the form ${}^nC_k : {}^nC_{k+1} : {}^nC_{k+2}$. The sum of all possible values of (n + k) is _____.

86. The mean and standard deviation of 6 observations are 8 and 4 respectively. If each observation is multiplied by 3, the new standard deviation of the resulting observations is _____.

87. The value of $\lim_{x\to 0} \frac{\tan x\sqrt{\tan x} - \sin x\sqrt{\sin x}}{x^3.\sqrt{x}}$ is _____.

88. Three people each flip two fair coins. The probability that exactly two of the people flipped one head and one tail, is _____.

89. If $P(S)$ denotes the set of all subsets of a given set S, then the number of one-to-one functions from the set $S = \{1, 2, 3\}$ to the set $P(S)$ is _____.

90. A triangle ABC satisfies the relation $2\sec 4C + \sin^2 2A + \sqrt{\sin B} = 0$ and a point P is taken on the longest side of the triangle such that it divides the side in the ratio 1 : 3. Let Q and R be the circumcentre and orthocentre of Δ ABC. If PQ : QR : RP = 1 : α : β, then the value of $\alpha^2 + \beta^2$ is _____.